

A Grid of Numbers · AS · Answers

U09b · Student copy · 24 written + 6 pen-first = 30 marks · Fresh authored questions

Mark points, not lines. [1] = one point; Max = cap; TV = too vague; NE = not enough. Accept equivalent correct explanations and calculation methods. Where a calculation has a method point and an answer point, a bare correct answer earns the answer point only; do not award one mark per working line.

1. Samples and grids [4]

- (a) Bits used to represent each pixel's colour [1]. (b) $2^6 = 64$ [1].
(c) Image resolution: dimensions/count of pixels in the image [1]; screen resolution: dimensions/count of display pixels [1]. Physical size in inches alone is not resolution.

2. Decimal and binary units [4]

- (a) $900 \times 600 \times 4 \div 1\,000\,000$ [1] = **2.16 MB** [1]. No $\div 8$.
(b) $768 \times 512 \times 16 \div 8 \div 1024^2$ [1] = **0.75 MiB** [1]. Do not interchange MB and MiB.

3. Header and payload [4]

- (a) Any two distinct items [1 each, Max 2]: colour depth; format identifier; compression method; pixel-data offset. Accept other valid format metadata, including a declared file/data size. Width/height are excluded by the stem.
(b) $128 + 4096 = 4224$ bytes [1]; compression changes the stored payload length [1]. The 128-byte header belongs to the stated model; actual formats vary.

4. Vector recipe [4]

- (a) Circle or rectangle [1]; a valid non-colour property of that object, e.g. circle radius/centre, rectangle width/height/position [1].
(b) Ordered list of drawing objects and their properties [1]; swapping makes the rectangle appear over the circle at the overlap [1]. "It changes the picture" is too vague.

5. Justify from the task [4]

- (a) Bitmap [1]; pixel samples suit the many irregular colour/detail variations in the photograph [1].
(b) Vector [1]; simple geometric objects can be rendered at different sizes without enlarging a fixed source pixel grid [1]. Accept a comparably concrete task-based reason. "Better quality" / "always smaller" alone earns no reason mark. These are suitable choices, not claims that alternatives are impossible.

6. Changes multiply [4]

- (a) $2 \times 3 \times (16 \div 8)$ [1] = **12 times** [1].
(b) More representable colours can retain finer colour differences / smoother gradients at capture [1]; changing an old encoding cannot by itself recover previously discarded differences [1].

7. Pen first [6]

- (a) Read bits in order and map 1 to black / 0 to white [1]; move to a new row after every four bits [1]. Accept equivalent numbered steps.
(b) Only the top-left to bottom-right diagonal is black. One mark for rows 1–2 both correct; one for rows 3–4 both correct.
(c) $12 \rightarrow 0C$; $128 \rightarrow 80$; $225 \rightarrow E1$. Award [1] for all three channel values correct, allowing C for 12 at this stage; [1] for the correctly ordered, fixed-width result **#0C80E1**. Accept lowercase and omission of #; do not accept a five-digit result for the width mark.

Above and beyond · unmarked. Both patches have two 1s out of four samples, giving **0.5**. This one output cannot distinguish the two different originals: averaging discarded their arrangement. It does not prove that no useful enhancement is ever possible.